NAME:__________________________ PDG:_________ Register No: ____

ANDERSON JUNIOR COLLEGE

JC 2 Preliminary Examination 2012

CHEMISTRY **8872/02**

Higher 1 **13 September 2012**

Paper 2 **2 hours**

Candidates answer Section A on the Question Paper.

Additional Materials: Data Booklet
Writing paper

READ THESE INSTRUCTIONS FIRST

Write your name, PDG and register number on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **two** questions on separate writing paper.
Start each question on a fresh sheet of paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

<table>
<tr><th colspan="7">For Examiner's Use</th></tr>
<tr><th rowspan="3">Paper 1 (33%)</th><th colspan="6">Paper 2 (67%)</th></tr>
<tr><th colspan="3">Section A</th><th colspan="2">Section B</th><th rowspan="2">Total</th></tr>
<tr><th>Q1</th><th>Q2</th><th>Q3</th><th></th><th></th></tr>
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This document consists of **12** printed pages.

Section A

Answer **all** the questions in this section in the spaces provided.

1 Pure iron is soft but it can be significantly hardened and strengthened by adding impurities, such as carbon, from the smelting process. This results in the formation of different iron alloys which find many industrial uses due to their great range of desirable properties and the abundance of iron.

Potassium manganate(VII) is commonly used in quantitative analyses of iron in the laboratory. A typical procedure involves the reaction of the manganate(VII) ion and aqueous Fe^{2+} ions in an acidic solution. The manganate(VII) ion reacts as follow:

$$MnO_4^- + 8H^+ + 5e \longrightarrow Mn^{2+} + 4H_2O$$

(a) **(i)** Write a half–equation for the reaction of Fe^{2+} ions.

....

(ii) What is the overall equation for the reaction of manganate(VII) ion and Fe^{2+} ions?

....

(iii) Determine the change in oxidation number of iron and manganese in the reaction.

....

....

(iv) The following analysis was carried out to determine the purity of iron in carbon steel, a metal alloy made up of iron and carbon.

A 1.00 g sample of carbon steel was first reacted with acid to produce aqueous Fe^{2+} ion. The solution was then titrated with 0.20 mol dm^{-3} potassium manganate(VII).

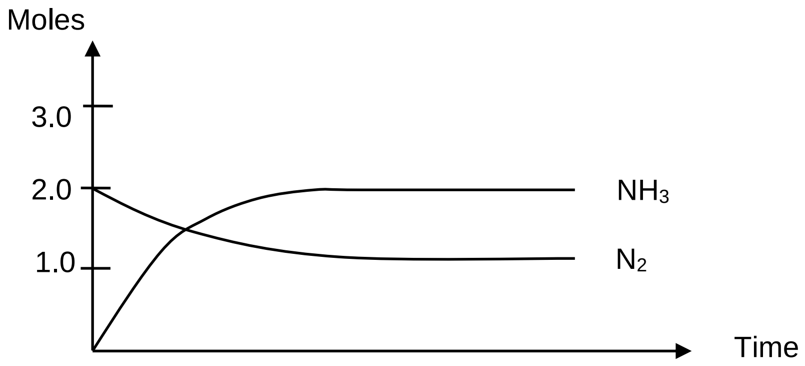
Assuming that carbon steel alloy contains a maximum carbon content of 2.1% by mass, determine the **minimum** volume of potassium manganate(VII) required to react with the iron in the alloy.

[5]

(b) In the Haber process, iron is used as a catalyst for the reaction of nitrogen and hydrogen to produce ammonia.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

A nitrogen–hydrogen mixture in the mole ratio of 1:4 was allowed to react in a 1 dm^3 closed vessel reactor at temperature T and the reaction reached an equilibrium with ammonia. The graph shows how the number of moles of nitrogen and ammonia varies with time.



(i) Determine the number of moles of hydrogen gas present in:

Initial mixture: ....

Equilibrium mixture: ....

(ii) Hence, calculate the value of K_c, giving its units.

(iii) State and explain how the position of equilibrium might change if the volume of the vessel containing the equilibrium gaseous mixture suddenly increases.

....

....

....

(iv) With the aid of a sketch of the Boltzmann distribution, explain how iron metal speeds up the attainment of equilibrium in the Haber process.

....

....

....

.... [8]

[Total: 13]

2 **(a)** **(i)** Complete the table given below using the values in the *Data Booklet*.

Ion	**Ionic radius / nm**
P^{3-}	
S^{2-}	
Cl^{-}	

(ii) Suggest an explanation for the trend in the ionic radii of phosphorus, sulfur and chlorine.

....

....

.... [2]

(b) The hydrogen halides decompose on heating as shown.

$$2HX\text{ (g)} \rightleftharpoons H_2\text{(g)} + X_2\text{(g)} \quad (X = Cl, Br \text{ or } I)$$

(i) Explain, with the aid of a balanced equation, what is meant by the term *bond energy of chlorine*.

....

....

(ii) Suggest how the extent of reaction will vary from C*l*, Br to I.

....

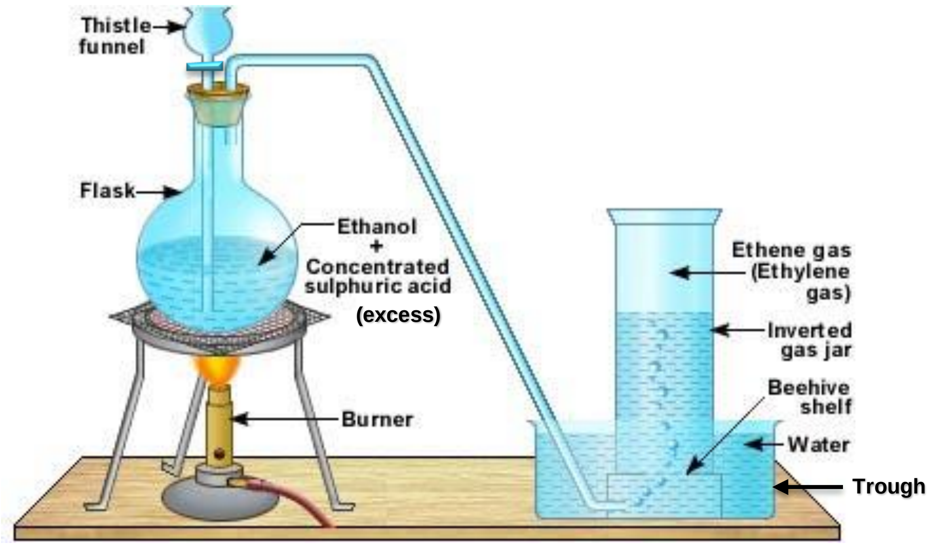
(iii) Use the bond energy values from the *Data Booklet* to calculate ΔH for this reaction when X = C*l* and when X = I.

(iv) Use your answers from **(b)(iii)** to explain the trend you have stated in **(b)(ii)**.

....

.... [5]

(c) Alkenes can be prepared in the laboratory by heating alcohols with **excess** concentrated sulfuric acid. The apparatus shown below can be used to prepare a sample of ethene.



Source: http://educationalelectronicsusa.com/c/org_chem–v.htm

(i) Write an equation for the reaction that occurs.

....

The ethene collected can be further purified by first bubbling it through a solution **A** and then passing it through a test tube containing anhydrous calcium chloride.

(ii) Suggest an identity for solution **A** and explain its purpose.

....

(iii) Suggest why anhydrous calcium chloride is required to obtain pure ethene.

....

Ethane–1,2–diol may be formed instead of ethene if the water in the trough is replaced with reagent **B**.

(iv) What is the identity of reagent **B**?

....

(v) What changes do you expect to observe in the trough?

....

(vi) Suggest one simple chemical test that could be used to distinguish between ethane–1,2–diol and ethanol, and state the observation expected for each compound.

....

....

....

.... [7]

[Total:14]

3 From a scientific point of view, coffee is an incredibly complex substance. It is estimated to contain more than 2000 chemical constituents, and among these are organic acids which contribute to the perceived acidity of coffee.

The acids found in roasted coffee may be classified into three groups: aliphatic, chlorogenic and alicyclic carboxylic acids, and phenolic acids. The table below shows the concentrations of three carboxylic acids extracted by brewing a commercial roasted coffee sample at 94°C for 5 minutes and then cooled to 25°C.

Acids present in coffee sample	pK_a at 25°C	Concentration in mg per litre at 25°C
Methanoic acid, HCO_2H	3.75	93.02
Ethanoic acid, CH_3CO_2H	4.75	242.67
Lactic acid, $CH_3CH(OH)CO_2H$	3.08	109.67

(a) **(i)** Write an expression for the acid dissociation constant, K_a, for ethanoic acid.

(ii) State how the acid strength of ethanoic acid compares with that of methanoic acid and lactic acid.

....

....

(iii) What is the concentration, in mol dm^{-3}, of ethanoic acid present in the sample?

(iv) If ethanoic acid is the only acid present in the coffee sample, calculate the molar concentration of hydrogen ions contributed by ethanoic acid if the coffee sample has pH of 3.6.

(v) The actual molar concentration of hydrogen ions contributed by ethanoic acid in the coffee sample is, in fact, different from the calculated value in **(a)(iv)**.

Explain how and why this is so.

....

....

.... [6]

(b) The hydrangea is often called "Nature's litmus" because its flowers change colour depending on the pH value and mineral content of the soil in which it grows.

To ensure sufficient acidity for a blue bloom, the pH of the flowerbed should be kept at 5.0 – 5.2. This can be achieved by applying a solution of aluminium sulfate.

Other means of providing an acidic medium include adding a mulch of pine needles, oak leaves or coffee grounds to the flowerbed. Acids from these mulches leach into the soil, helping to produce blue blooms.

(i) With the aid of an appropriate equation, explain how an aqueous solution of aluminium sulfate is used to provide an acid medium for the cultivation of blue hydrangea.

....

....

....

(ii) A gardener wishes to prepare a flowerbed suitable for cultivating blue hydrangea. However, he only has a 800 cm^3 sample of aqueous sulfuric acid of pH 4.8.

What is the volume of water that needs to be added to the sulfuric acid in order to raise its pH to 5.1 before using it on the flowerbed?

[3]

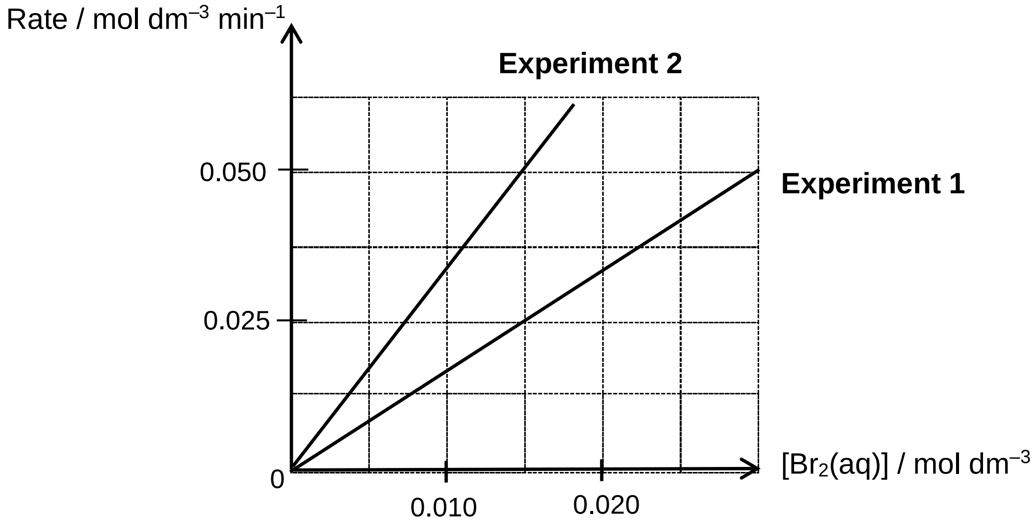
(c) Methanoic acid can react with bromine according to the following equation.

$$HCO_2H(aq) + \underset{\textit{orange brown}}{Br_2(aq)} \rightarrow 2H^+(aq) + \underset{\textit{colourless}}{2Br^-(aq)} + CO_2(g)$$

Two separate experiments were conducted to study the kinetics of the above reaction in the presence of excess methanoic acid.

Experiment	Initial $[HCO_2H]$ / mol dm^{-3}
1	0.50
2	1.00

The experimental results were represented in the graph as shown.



(i) Why is an excess of methanoic acid used in each experiment?

....

....

(ii) With the aid of the graph, determine the order of reaction with respect to aqueous bromine.

Order : ....

Explanation : ....

(iii) How does the rate of **Experiment 1** compare with that for **Experiment 2** when the concentration of aqueous bromine was 0.015 mol dm^{-3}?

....

(iv) Hence, deduce the rate equation for the reaction between methanoic acid and aqueous bromine.

.... [4]

[Total: 13]

Section B

Answer **two** questions from this section on separate writing paper.

4 Phosphorus and sulfur form many useful chlorides and oxychlorides.

Phosphorus pentachloride, PCl_5, is one of the most important chlorides, others being PCl_3 and $POCl_3$. PCl_5 exists as white solid and is widely used as a chlorinating agent of organic compounds.

Thionyl chloride, $SOCl_2$ and sulfuryl chloride, SO_2Cl_2 are both colourless liquids at room temperature. The properties of these two sulfur oxychlorides are quite different; thionyl chloride is a source of chloride ions whereas sulfuryl chloride is a source of chlorine.

(a) In the solid state, PCl_5 has an ionic lattice structure consisting of PCl_x^+ and PCl_6^- ions.

Deduce a likely identity and the shape of the cation, PCl_x^+, and draw its dot–and–cross diagram.

[2]

(b) The phosphorus chlorides and sulfur oxychlorides react with water to produce acidic compounds.

(i) PCl_5 reacts with water in two separate ways.

- When a few drops of water are added to the solid, white fumes are evolved and a white solid remains which is insoluble in water.
- When large amount of water is added to the solid, a very acidic solution results.

Write equations, including state symbols, for these two reactions and explain the observations.

(ii) Both $SOCl_2$ and SO_2Cl_2 react vigorously with water.

When 1 mol each of $SOCl_2$ and SO_2Cl_2 is added separately to water, 2 mol of HCl and a sulfur–containing product are formed in each reaction. The sulfur in the two compounds have different oxidation states.

Suggest the identities of the sulfur–containing products for each reaction and state the oxidation number of sulfur in each case.

[5]

(c) The decomposition of SO_2Cl_2, forming sulfur dioxide and chlorine gas, is a first order reaction. The reaction has a half–life of 245 min at 600 K.

If 3.6×10^{-3} mol of SO_2Cl_2 in a flask is allowed to decompose, how long will it take approximately for the amount of SO_2Cl_2 to decrease to 2.3×10^{-4} mol?

[1]

Four organic compounds, **G**, **H**, **J** and **K**, each have the empirical formula CH_2O. The numbers of carbon atoms in their molecules are shown in the table.

Compound	Number of carbon atoms
G	1
H	2
J	3
K	4

In **H** and **J**, the carbon atoms are bonded directly to one another.

G gives a red precipitate when treated with Fehling's reagent.

H and **J** give a brisk effervescence with Na_2CO_3(aq).

(d) **(i)** Draw the structural formulae for the compounds represented by **G** and **H**.

(ii) There are two possible structures for **J**.

Suggest **two** different simple chemical tests to differentiate **H** and the two structures of **J**. State the observation expected for each compound in the tests.

(iii) **G** reacts with lithium aluminium hydride to form a compound **L**.

Identify **L** and suggest how you can confirm the functional group present in **L** using $SOCl_2$.

[8]

(e) A possible identity for **K** is butenedioic acid, $HO_2CCH{=}CHCO_2H$, a colourless crystalline solid used in textile processing and as an oil and fat preservative.

Describe what you might observe, and give the structural formulae of the products, when butenedioic acid is

(i) heated with hydrogen gas in the presence of finely–divided nickel in a closed vessel connected to a pressure gauge, and

(ii) shaken with aqueous bromine.

[4]
[Total:20]

5 **(a)** Sketch a graph showing the variation of melting point for the elements from Na to Ar. Explain the shape of the graph in terms of the structures and bonding of the elements.

[4]

(b) Like carbon, silicon forms a homologous series of inorganic compounds, analogous to alkanes. They consist of only hydrogen and silicon atoms bonded together by single bonds, forming linear saturated compounds.

Disilane, Si_2H_6, and ethane have similar structure. Both react with oxygen to produce the dioxide and water, but disilane is much more unstable. In the presence of air, disilane rapidly and spontaneously oxidises to SiO_2 and H_2O.

(i) The enthalpy change of combustion of ethane can be determined using the energy cycle as shown.

$$CH_3CH_3\,(g) \;+\; {}^7/_2\,O_2\,(g) \longrightarrow 2CO_2(g) \;+\; 3H_2O(l)$$

$$2C\,(s) \;+\; 3H_2\,(g) \;+\; {}^7/_2O_2\,(g) \longrightarrow 2CO_2\,(g) \;+\; 3H_2\,(g) \;+\; {}^3/_2O_2\,(g)$$

Copy the energy cycle onto the writing paper and indicate the following three values of enthalpy changes at the appropriate arrows.

$\Delta H_f^\ominus$ (ethane) $= -84$ kJ mol^{-1}
$\Delta H_f^\ominus$ (carbon dioxide) $= -394$ kJ mol^{-1}
$\Delta H_f^\ominus$ (water) $= -285$ kJ mol^{-1}

Hence, calculate the standard enthalpy change of combustion of ethane.

(ii) With the aid of the *Data Booklet*, suggest reasons for the comparative instability of disilane and its ready reaction with air.

[5]

(c) In a recent study carried out at an isolated site in the Colorado mountains, compound **P**, with molecular formula $C_5H_{10}O$, was found to be the most abundant volatile organic compound present in the atmosphere.

Analysis on **P** reveals the following information:

- **P** shows no reaction with 2,4–dinitrophenylhydrazine but liberates a gas when sodium metal is added to it.
- Heating **P** with acidified potassium manganate(VII) produces compound **Q**, $C_4H_8O_3$, and colourless gas, **R**.
- **Q** liberates gas **R** with aqueous sodium carbonate solution.
- Like **P, Q** shows no reaction with 2,4–dinitrophenylhydrazine.

Deduce the identities for the compounds **P**, **Q** and gas **R** and explain the chemistry of the reactions involved.

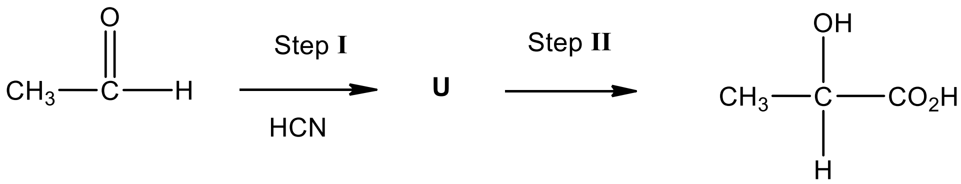
[11]
[Total:20]

6 **(a)** Sketch a graph showing the variation of electrical conductivity for the elements from Na to Ar. Explain the shape of the graph in terms of the structures and bonding of the elements. [4]

(b) A mixture of lactic acid and its sodium salt forms an acidic buffer.

Using this mixture as an example, illustrate the principles of how acidic buffer solutions work. [4]

(c) Lactic acid can be synthesized from ethanal in a reaction route as shown.



(i) Name the type of reaction that occurs at Step **I** and draw the structural formula of **U**.

(ii) Suggest reagent and condition required for Step **II**.

(iii) Traces of an impurity, $C_6H_8O_4$, are found together with the final product. Suggest how this impurity is formed and deduce its structural formula.

(iv) A study was carried out to determine the relative rates of the reaction of ethanal with HCN in step **I** under different conditions.

The following set of results was obtained.

Condition	*relative rate*
aqueous solution	slow
acidified solution	virtually zero
alkaline solution	very rapid

What conclusion can be drawn from these data about the species involved in the reaction? Explain your answer.

[7]

(d) 2–chloropropanoic acid, $CH_3CH(Cl)CO_2H$, can be converted into lactic acid under suitable conditions in the laboratory.

(i) State the reagent and condition required for the conversion.

(ii) State and explain the relative acidities of 2–chloropropanoic acid and propanoic acid. [5]

[Total: 20]